

Research Assistant/Postdoctoral Research Assistant

About the Role

The School of Electronic Engineering and Computer Science at Queen Mary University of London is looking to recruit a Postdoctoral Research Assistant to support the projects led by **Dr. Ahmed Sayed**. The main project (KUber) is, EPSRC-funded, aiming at building novel distributed architectures to facilitate the exchange and delivery of acquired knowledge among machine learning entities at scale. For brief details of projects including KUber see (<https://sayed-sys-lab.github.io/projects/>).

About You

Applicants should preferably have an MSc/PhD in Computer Science/Engineering. This is mainly aimed at PhD graduates or PhD students nearing their PhD completion. They should have expertise in distributed systems and computer networks. Good knowledge and practical skills in distributed machine learning, networked systems, data management, and/or software engineering, are recommended for this position. The post requires strong programming and system development abilities. Prior involvement in notable open-source projects is a plus. Applicants should have publications in leading conferences and journals in the aforementioned fields.

Benefits

The post is based at the Mile End Campus in London. It is a full-time, fixed-term appointment with a tentative start date between 1 Jan-Apr 2024 for up to 36 months. The starting salary will be Grade 4, in the range of £36,572 - £47,148 per annum dependent on qualifications and experience, inclusive of London Allowance.

Informal enquiries should be addressed to **Dr. Ahmed Sayed** (www.eecs.qmul.ac.uk/~ahmed) at ahmed.sayed@qmul.ac.uk

Formal Applications by **10 Jan 2024 23:55 GMT**: <https://qmul-jobs.tal.net/vx/candidate/apply/184>

Some Relevant Recent Publications

Conferences:

A. M. Abdelmoniem, AN Sahu, M Canini, SA Fahmy. "REFL: Resource Efficient Federated Learning". In Proceedings of ACM **EuroSys**, Rome, Italy, 2023
Atal Sahu, Aritra Dutta, A. M Abdelmoniem, Trambak Banerjee, Marco Canini, Panos Kalnis. Rethinking gradient sparsification as total error minimization. Proceedings of **NeurIPS**, Spotlight (Top 3%), 2022.
A. M. Abdelmoniem, M. Canini. "DC2: Delay-aware Compression Control for Distributed Machine Learning". IEEE Computer Communications Conference (**INFOCOM**), May 2021.
A. M. Abdelmoniem, A. Elzanaty, M. Alouni, M. Canini "An Efficient Statistical-based Gradient Compression Technique for Distributed Training Systems". Proceedings of the International Conference on Machine Learning and Systems (**MLSys**), 2021

Journals:

Z. Tian, Z. Wang, A. M. Abdelmoniem, G. Liu and C. Wang, "Knowledge Representation of Training Data With Adversarial Examples Supporting Decision Boundary", IEEE Transactions on Information Forensics and Security (**IEEE TIFS**), 2023.
A. M. Abdelmoniem, Chen-yu Ho, Pantelis Papageorgiou, Marco Canini. "A Comprehensive Empirical Study of Heterogeneity in Federated Learning". IEEE Internet of Things Journal (**IEEE IoTJ**), 2023.
A. M. Abdelmoniem and B. Bensaou, "Enhancing TCP via Hysteresis Switching: Theoretical Analysis and Empirical Evaluation," in IEEE/ACM Transactions on Networking (**IEEE ToN**), 2023.
A. J. Abu, B. Bensaou, A. M. Abdelmoniem, "Dimensioning the pending interest table in content-centric networks". Future Generation Computer Systems (**FGCS**), Elsevier, 2023

Patent:

Statistical-based gradient compression method for distributed training system, WO2022003562A1